MATLAB Random Bit Error Rate Simulator

DIGITAL COMMUNICATIONS SIMULATION REPORT & AWGN CHANNEL PERFORMANCE ANALYSIS

1. EXECUTIVE SUMMARY & OBJECTIVE

This technical report presents a full simulation and mathematical analysis of a digital communication link modeled in MATLAB. The core objective is to measure the **Bit Error Rate (BER)** of binary data transmitted over an **Additive White Gaussian Noise (AWGN)** channel across four target Signal-to-Noise Ratio (SNR) levels: **0 dB, 3 dB, 6 dB, and 12 dB**.

The simulation transmits **N = 10,000 random binary bits** using Binary Phase Shift Keying (BPSK) antipodal voltage mapping (+1 V / -1 V). Corrupted signals are decoded at the receiver using a zero-threshold decision detector to quantify data corruption.

2. MATHEMATICAL FRAMEWORK & SYSTEM ARCHITECTURE

System Mathematical Equations

A. Antipodal Signal Mapping: Binary sequence $b[n] \in \{0, 1\}$ with equal probability $P(0)=P(1)=0.5$ maps to voltage amplitudes $s[n] = A \cdot (1 - 2 \cdot b[n])$ (Bit 0 → +1V, Bit 1 → -1V for amplitude A = 1).

B. AWGN Channel & Noise Variance: Zero-mean Gaussian noise $w[n] \sim N(0, \sigma_w^2)$ is added to signal $s[n]$. SNR conversion: $\mathbf{SNR_{linear} = 10^{(SNR_{dB} / 10)}}$. Noise variance: $\mathbf{\sigma_w^2 = A^2 / (2 \cdot SNR_{linear})}$. Received signal: $r[n] = s[n] + w[n]$.

C. Zero-Threshold Bit Detector: Detected bits $b\blacksquare[n] = 1 \text{ if } r[n] < 0, \text{ else } 0$. Bit error rate: **BER = Total Errors / N**.

3. MATLAB SIMULATION CODE ARCHITECTURE

```
clc; clear; close all;
N = 10000; A = 1; SNR_dB = [0 3 6 12];
bits = randi([0 1], 1, N);             % Generate N uniform random bits (0 or 1)
tx = A * (1 - 2 * bits);                 % Map bits: 0 -> +1V, 1 -> -1V
BER_values = zeros(size(SNR_dB));

for k = 1:length(SNR_dB)
    snr_db = SNR_dB(k);
    snr_linear = 10^(snr_db / 10);      % Convert SNR dB to linear scale
    noise_var = A^2 / (2 * snr_linear);  % Calculate noise variance
    noise = sqrt(noise_var) * randn(1, N);% Generate zero-mean Gaussian noise
    rx = tx + noise;                     % Add noise to transmitted signal
    bits_hat = rx < 0;                   % Zero-threshold bit detection
    errors = bits ~= bits_hat;           % Identify corrupted bits
    BER_values(k) = sum(errors) / N;     % Calculate Bit Error Rate
end
semilogy(SNR_dB, BER_values, 'o-', 'LineWidth', 1.5); % Logarithmic BER plot
```

- **randi([0 1], 1, N):** Generates N uniform equiprobable random binary digits (0 or 1).
- **tx = A * (1 - 2 * bits):** Performs BPSK antipodal mapping (Bit 0 → +1V, Bit 1 → -1V).
- **noise_var = A^2 / (2 * snr_linear):** Computes exact Gaussian noise variance for target SNR.
- **rx < 0:** Implements zero-voltage threshold decision rule for bit recovery.

4. SIMULATION WAVEFORMS & SIGNAL ANALYSIS

Figure 1 through Figure 3 showcase the first 100 samples of binary bit generation, transmitted antipodal electrical signals, and noisy received signals at 0 dB SNR.

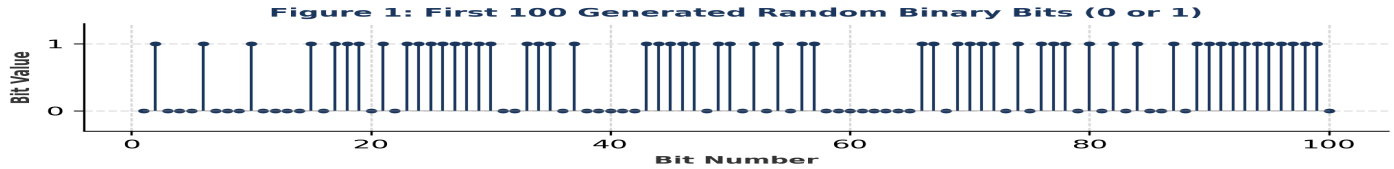


Figure 1: First 100 discrete random binary bits generated using uniform integer distribution randi([0 1]).

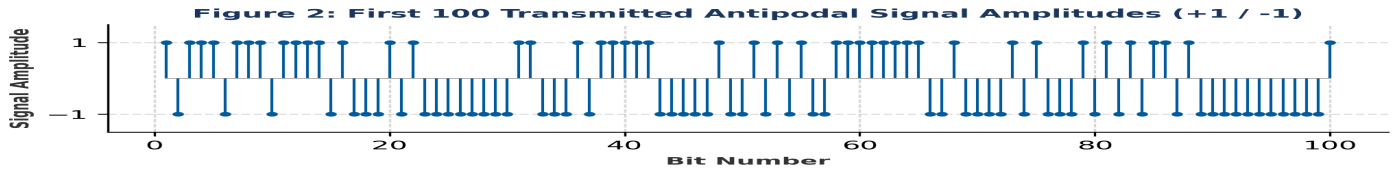


Figure 2: Transmitted antipodal signal waveform (+1 V for bit 0, -1 V for bit 1).

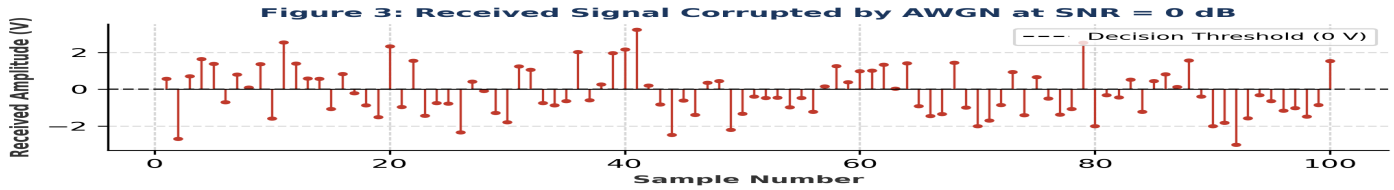


*Figure 3: Received signal corrupted by Gaussian noise at **0 dB SNR** ($\sigma_w^2 = 0.500$). Threshold crossings at 0 V cause bit errors.*

5. PERFORMANCE EVALUATION: BER VS. SNR RESULTS

Quantitative simulation results across 10,000 transmitted bits are compiled in Table 1 and depicted on the semi-logarithmic BER curve in Figure 4.

SNR (dB)	SNR (Linear)	Noise Variance (σ■²)	Corrupted Bits	Bit Error Rate (BER)	Error %
0 dB	1.000	0.5000	756	0.0756	7.56%
3 dB	1.995	0.2506	221	0.0221	2.21%
6 dB	3.981	0.1256	17	0.0017	0.17%
12 dB	15.849	0.0315	0	0.0000	0.00%

Table 1: Simulated Bit Error Rate (BER) and noise variance metrics across 10,000 transmitted bits.

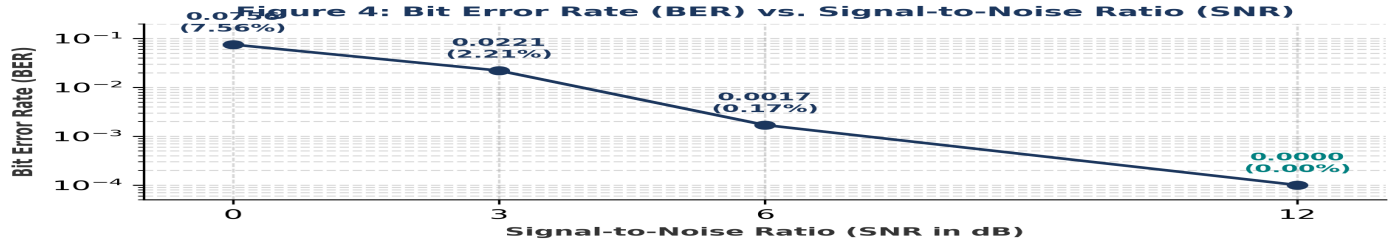


Figure 4: Bit Error Rate (BER) vs. Signal-to-Noise Ratio (SNR in dB) on a semi-logarithmic scale.

6. ENGINEERING DISCUSSION & KEY FINDINGS

Key Analytical Takeaways

Exponential Error Reduction: Increasing SNR from 0 dB to 12 dB reduces noise variance from 0.500 to 0.0315, driving BER down from 7.56% to 0.00% across N=10,000 bits.

Threshold Inversion Dynamics: At 0 dB SNR ($\sigma_w = 0.707$ V), large Gaussian noise spikes frequently cross the 0 V decision threshold, causing frequent decision errors.

Practical Channel Standards: Modern communication links rely on target SNRs ≥ 10-12 dB or Forward Error Correction (FEC) codes to maintain low BER (< 10^{-5}) required for data integrity.